

Enhancing biodiesel yield through disc-surface texturing in a spinning disc reactor: Experimental and CFD analysis

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ABSTRACT

Biodiesel is a sustainable alternative to conventional fossil fuels. However, mass transfer resistance remains a major challenge in its production. In this study, continuous biodiesel production from refined palm oil was investigated in a spinning disc reactor (SDR) using disc-surface texturing as a passive intensification strategy, supported by computational fluid dynamics (CFD)-based hydrodynamic analysis. Experiments were performed at 60 °C with a total feed flow rate of 270 mL/min, a methanol-to-oil molar ratio of 6:1, and a NaOH loading of 0.75 wt.% of oil. Three disc surfaces (smooth, P36 and P24 roughness) were tested at rotational speeds of 500, 1000 and 1500 rpm. The results showed that disc roughness significantly enhanced reactor performance, with FAME yield increasing from 84.6% (smooth) to 90.0% (P36) and 97.3% (P24) at 1000 rpm. CFD-derived methanol volume fraction and TKE results indicated more persistent spiral phase structures and enhanced turbulence-related hydrodynamic activity on roughened discs, supporting the hydrodynamic interpretation of the observed FAME yield trends. Energy-normalized benchmarking with various intensified reactors showed that the roughened P24 achieved a high yield efficiency of 1.402×10^{-3} g/J, demonstrating competitive continuous biodiesel production with ultra-short contacting times.

1. Introduction

Renewable and clean energy sources have become increasingly important due to growing interest in sustainable energy solutions. Biodiesel, a renewable biofuel, has received considerable attention as a promising clean-energy alternative because of its high heating value, clean combustion properties, low environmental impact, and capability to directly replace petroleum diesel [1–3]. Typically, biodiesel is produced through the transesterification of triglycerides derived from animal fats or vegetable oils with alcohol, commonly methanol, resulting in fatty acid methyl esters (FAMES) and glycerol [4,5]. The transesterification is commonly catalyzed by homogeneous catalysts, such as

alkaline catalysts (NaOH, KOH and CH_3ONa) and acid catalysts (H_2SO_4 , H_3PO_4 and *p*-toluene sulfonic acid), or heterogeneous catalysts, including metal oxides, sulfated zeolites, and hydrotalcites [6–8]. NaOH and KOH are commonly used homogeneous alkaline catalysts for biodiesel production from low free fatty acid (FFA) feedstocks [9,10]. However, alkaline hydroxide catalysts are sensitive to FFA and water, which may promote saponification, soap formation, and reduced apparent FAME yield [11,12]. Compared with KOH, NaOH has been reported in some systems to be associated with greater soap formation and yield loss under unfavorable feedstock or reaction conditions [13]. Nevertheless, NaOH remains widely used for low FFA oils because it is inexpensive, readily available, and effective under mild transesterification conditions [9,10]. Therefore, NaOH was selected in this

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Nomenclature*Symbols*

$\sum A$	Sum of the GC peak areas for C14:0 to C24:1
A_{EI}	Peak area of methyl heptadecanoate standard
C_{EI}	Concentration of methyl heptadecanoate solution (mg/mL)
D	Disc diameter (cm)
F_{oil}	Molar flow rate of oil (mol/min)
F_{σ}	Surface tension force (N/m ³)
F_1	Blending function value
g	Gravitation force (m/s ²)
h	Liquid film thickness (mm)
h_N	Liquid film thickness prediction from Nusselt model (mm)
m	Mass of biodiesel sample (mg)
m_{FAME}	Mass of FAME produced during the steady-state sampling period
N	Nusselt model
p	Pressure (Pa)
P	Average power consumption (W)
Q	Total volumetric flow rate (mL/min)
r	Position along the disc radius (cm)
R	Rotational disc speed (rpm)
r_i	Radius of liquid feeding pipe (cm)
r_s	Radius of synchronization zone (cm)
t_s	Steady-state sampling time (s)
TKE	Turbulent kinetic energy (J/kg)
u	Mean fluid velocity (m/s)
V_{EI}	Volume of methyl heptadecanoate solution (mL)
V_R	Reactor volume (mL)
Y_{FAME}	FAME yield (%)
Y^+	Dimensionless wall distance used to assess near-wall mesh resolution

Greek symbols

α	Volume fraction of the (q) phase
ε	Turbulent dissipation rate (W/kg)
μ	Dynamic viscosity (Pa.s)
ν	Kinematic viscosity (m ² /s)
Ω	Angular velocity of the rotating disc (rad/s)

ω	Specific dissipation rate (1/s)
ϕ	Indicator variable for grid independence analysis
ρ	Density (kg/m ³)
τ	Residence time (s)

Subscripts and Superscripts

g	Gas phase
i	Initial condition or node index
l	Liquid phase
q	Any phase (gas or liquid)
s	Steady-state condition
T	Turbulence

Acronyms

E_k	Ekman number
CFD	Computational Fluid Dynamics
CFL	Courant–Friedrichs–Lewy
FAME	Fatty acid methyl ester
FFA	Free fatty acid
GC	Gas chromatography
HC	Hydrodynamic cavitation
HSM	Homogenizer
LES	Large Eddy Simulation
MBR	Multiple baffle reactor
MCR	Microchannel reactor
MeOH	Methanol
MS	Mechanically stirred reactor
QUICK	Quadratic Upstream Interpolation for Convective Kinematics
RANS	Reynolds-Averaged Navier–Stokes
RD	Reactive distillation
RPO	Refined palm oil
RTD	Residence time distribution
RTR	Rotating tube reactor
SDR	Spinning disc reactor
SIMPLE	Semi-implicit method for pressure-linked equations
SPW	Spiral pulse-like waves
SSW	Spiral sine-like waves
SST	Shear stress transport
VOF	Volume of fluid

work for refined palm oil transesterification.

However, one of the major limitations encountered in conventional FAME production processes is high mass transfer resistance caused by immiscibility between the two liquid phases of oil and methanol. This limitation results in slow reaction rates, prolonged processing times, and increased production costs [14,15]. To address this issue, intensified reactors that significantly enhance mass transfer and mixing efficiency have been developed, such as hydrodynamic cavitation reactor (HC), rotating tube reactor [16], reactive distillation (RD) [17], microreactor [18], ultrasonic reactor [18], and microwave reactor [19]. In particular, microreactors and structured microchannels have also been widely investigated for intensified biodiesel production because their small channel dimensions and designed geometries enhance oil-alcohol contact, local shear, and mass-transfer rates [20,21]. For example, Yusuf et al. [22] reported that microchannel geometry strongly affects biodiesel production, with T- and Y-shaped microreactors achieving 93.3% and 94.3% yield, respectively, at 3.28 s, while a Tesla-shaped design achieved 97.9% at 4.85 s under comparable NaOH-catalyzed conditions. Other configurations, including omega-, Tesla-, and zig-zag-shaped channels, have also been reported to induce flow disturbance or vortex formation and improve liquid–liquid mass transfer [20,21]. Nevertheless, microreactors are often limited by low single-channel

throughput, and scale-up generally requires numbering-up or parallelization [20,23].

As another intensified reactor option, the spinning disc reactor (SDR) has been investigated for improving interphase contact in biodiesel production [24–26]. The SDR operates by forming a thin film of liquid reactants on a rapidly rotating disc surface, creating high shear forces and substantially increasing the interfacial area between the reactants [27,28]. These hydrodynamic characteristics can mitigate oil-methanol mass transfer resistance and improve apparent transesterification performance under suitable operating conditions [29]. Biodiesel (FAME) yields as high as 97% have been reported within a very short residence time of approximately 2–3 s, which is much shorter than that typically required in conventional mechanically stirred reactors [19]. Compared with structured microchannels that often require complex internal geometries and numbering-up to increase throughput, SDRs offer a relatively simple rotating-disc configuration and are suitable for continuous operation.

Computational fluid dynamics (CFD) has been increasingly used to investigate SDR hydrodynamics because direct characterization of thin liquid films and interfacial structures at high rotational speeds is challenging [30]. Previous CFD studies have applied VOF- and LES-based approaches to analyze liquid-film thickness, wave formation, velocity

distribution and flow behavior in spinning disc systems [30,31]. Multiphase CFD has also been used to evaluate liquid–liquid flow behavior, disc-surface effects, mass-transfer performance, and micro-mixing-related quantities in SDRs [32]. Recent studies on grooved or modified disc surfaces further show that surface structure can influence wavelet formation and micro-mixing behavior [27,33]. However, CFD studies that combine hydrodynamic descriptors with experimental biodiesel yield data for surface-textured SDRs remain limited.

Although previous SDR studies have investigated operating parameters such as rotational speed, feed flow rate, residence time, catalyst loading, methanol-to-oil ratio, and reaction temperature, the specific influence of disc-surface texturing on thin-film hydrodynamics and biodiesel production performance remains less clearly understood [25, 26,29]. In particular, further clarification is needed on how surface-induced changes in flow behavior, phase distribution, residence time, and mixing-related descriptors are associated with experimentally observed FAME yield. Because direct experimental observation of liquid-film behavior is challenging at high rotational speeds, CFD was used to support the hydrodynamic interpretation of the experimental results [32]. Therefore, this research aims to evaluate the performance of a surface-textured SDR for continuous biodiesel production from refined palm oil via base-catalyzed transesterification. The baseline operating conditions, including a reaction temperature of 60 °C, a total feed flow rate of 270 mL/min, a methanol-to-oil molar ratio of 6:1, and a NaOH loading of 0.75 wt.% based on oil weight, were selected based on conditions commonly reported in previous studies on homogeneous alkaline transesterification of refined palm oil [26,29]. These conditions were kept constant, while the rotational speed and disc-surface texture were varied to focus on their effects on FAME yield and thin-film hydrodynamics. Smooth, P36 and P24 disc surfaces were experimentally compared, and CFD simulations were used to analyze liquid–liquid flow

patterns, film thickness, phase distribution, residence time and turbulence-related quantities. By integrating experimental FAME yield measurements with CFD-derived hydrodynamic descriptors, this work provides insight into how disc-surface characteristics modify SDR hydrodynamics and how these changes are associated with FAME yield.

2. Methodology

2.1. Materials

Refined palm oil (RPO, OLEEN brand) used in this study was sourced commercially in Thailand. Its molecular weight was approximately 870.2 g/mol. Methanol (MeOH) of 99.8% was obtained from Qrec, Newzealand. Heptane solvent with 99.7% purity, was procured from Carlo Erba Reagents. In addition, sodium hydroxide (NaOH), utilized as a catalyst with a purity level of 97% and methyl heptadecanoate used as the internal standard for gas chromatography (GC) analysis, were purchased from Sigma-Aldrich.

2.2. Experimental set-up

The experimental apparatus used in this study is illustrated in Fig. 1. The spinning disc reactor (SDR) system consisted primarily of a rotating stainless-steel disc (diameter 13 cm), driven by a motor for controlling rotational speeds and enclosed in a stationary cylindrical chamber. Three different disc surfaces were investigated: (1) smooth surface, (2) roughness P36 surface (prepared using metal sanding disc P36, average roughness height of approximately 538 μm), and (3) roughness P24 surface (prepared using metal sanding disc P24, average roughness height of approximately 764 μm) [34]. It should be noted that the surface roughness follows the order, P24 surface > P36 surface > smooth

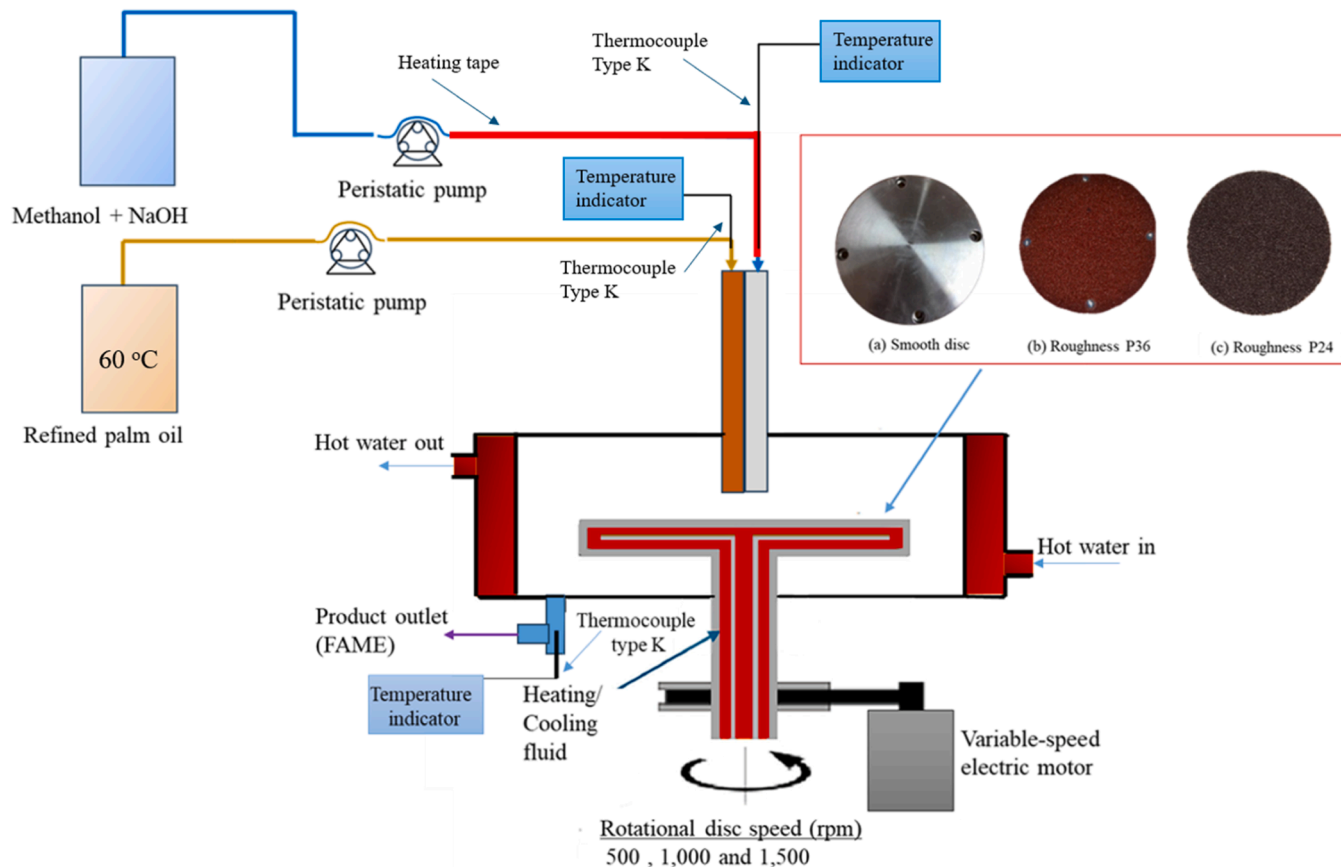


Fig. 1. Schematic diagram of the biodiesel production conducted in a SDR (modified from R. Sirisangsawang et al. [35]).

surface. Refined palm oil and a methanol solution containing NaOH catalyst were continuously supplied onto the disc surface by two separate peristaltic pumps. Both liquid feeds entered through two side-by-side inlet pipes (3 mm diameter), positioned 4.5 mm from the center of the rotating disc. The temperatures of both liquid feeds were monitored using thermocouples installed immediately before the reactor inlet and were maintained at 60 ± 2 °C before entering the rotating disc. The mixture of methanol and NaOH feed line was heated using heating tape, while the refined palm oil feed was thermally conditioned upstream to ensure stable pumping. The outlet temperature was also monitored using a thermocouple installed at the reactor outlet. The reactor temperature was precisely maintained at 60 °C using circulating hot water from a temperature-controlled water bath. Experiments were conducted at rotational disc speeds (R) ranging from 500, 1000 and 1500 rpm using three different disc surfaces, while maintaining a constant total feed flow rate of 270 mL/min ($Q_{RPO} = 214.4$ mL/min, $Q_{MeOH} = 55.6$ mL/min), a fixed methanol-to-oil molar ratio of 6:1, and a NaOH catalyst loading of 0.75 wt.% based on oil weight. The mean residence time reported in this study refers to the liquid film residence time on the rotating disc. It was calculated from the liquid holdup on the disc divided by the total volumetric flow rate, as expressed by $\tau = V_{\text{film}} / Q$, where V_{film} is the liquid holdup on the rotating disc and Q is the total feed flow rate. After reaching the disc edge, the liquid was immediately discharged from the rotating disc region and directed to the outlet, no intentional post-disc liquid holdup or recirculation was applied within the reactor chamber.

Samples were collected after allowing the system to reach steady-state operation (approximately 10 min). Upon collection, reaction progress was quenched immediately by placing samples into an ice bath. Subsequently, samples were centrifuged at 3000 rpm for 5 min to effectively separate glycerol from the biodiesel product (FAME). Each experimental condition was performed in duplicate, and the results are reported as mean values with error bars representing the standard deviation of duplicate measurements. The fatty acid methyl ester (FAME) content or biodiesel yield was analyzed by gas chromatography (GC) according to the EN 14,103 standard, as described in our previous study [26]. The FAME yield was calculated using Eq. (1) [36].

$$\text{FAME yield (\%)} = \frac{(\sum A) - A_{EI}}{A_{EI}} \times \frac{C_{EI} \times V_{EI}}{m} \times 100\% \quad (1)$$

where

$\sum A$ = Sum of the GC peak areas for C14:0 to C24:1
 A_{EI} = Peak area of methyl heptadecanoate standard
 C_{EI} = Concentration of the methyl heptadecanoate solution (mg/mL)
 V_{EI} = Volume of methyl heptadecanoate solution (mL)
 m = Mass of the biodiesel sample (mg)

In addition to FAME yield, the performance of the SDR was evaluated using yield efficiency. The yield efficiency was calculated as:

$$\text{Yield efficiency (g/J)} = \frac{m_{\text{FAME}}}{P \times t_s} \quad (2)$$

where m_{FAME} is the mass of FAME produced during the steady-state sampling period (g), P is the average power consumption measured using a power plug meter (W), and t_s is the steady-state sampling time (s). The measured power consumption included the electrical consumption of the spinning disc reactor, peristaltic pumps, feed-line heating tape, and water baths used for feed thermal conditioning and reactor temperature control.

In addition to yield efficiency, reactor productivity was also used as a volumetric performance indicator for comparison with other intensified reactors. The FAME productivity was calculated as:

$$\text{FAME productivity} = \frac{P_{\text{FAME}}}{V_R} \quad (3)$$

where P_{FAME} is the molar production rate of FAME (mol/min), V_R is the reaction volume, and the FAME productivity is expressed in mol/(mL·min) [37]. The molar production rate of FAME was calculated as:

$$P_{\text{FAME}} = 3 \times Y_{\text{FAME}} \times F_{\text{oil}} \quad (4)$$

where Y_{FAME} is the FAME yield expressed as a fraction, and F_{oil} is the molar flow rate of oil (mol/min). The factor of 3 accounts for the stoichiometric formation of three moles of FAME from one mole of triglyceride [37].

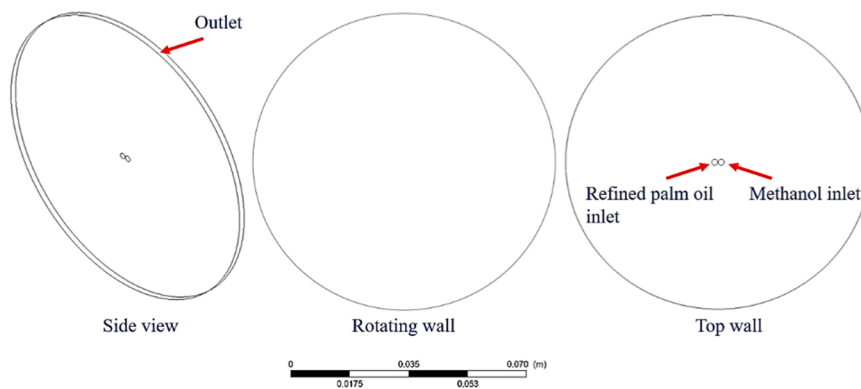
2.3. CFD simulations

2.3.1. Geometrical model and meshing

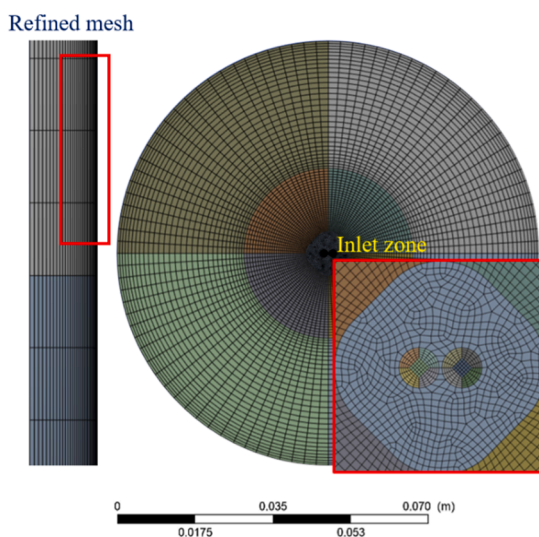
The SDR consists mainly of a motor, rotating disc, and reactor chamber. Its computational geometry was defined to replicate the experimental dimensions. In ANSYS Design Modeler (Fig. 2a), a flat cylindrical domain with 3 mm thickness was created. Three-disc surface conditions were modeled: (1) smooth stainless steel, (2) P36 produced with P36 sanding (≈ 538 μm average roughness), and (3) P24 produced with P24 sanding (≈ 764 μm average roughness). The P24 and P36 textured surfaces were represented using an equivalent sand-grain roughness approach [34]. This method approximates the effect of wall roughness on near-wall momentum transfer but does not explicitly resolve the actual groove geometry, local asperity distribution, or anisotropy of the textured surface. Therefore, the predicted roughness effect should be interpreted as an effective rough-wall approximation rather than a direct geometric representation of the real disc surface. Two circular inlet openings on the top face allow simultaneous entry of refined palm oil and methanol at the disc center (RPO and MeOH properties are listed in Table 1). Upon entry, both liquids spread into a thin film and flow outward in a spiraling pattern toward the rim. The bottom face of the cylinder represents the rotating disc surface. All remaining faces were defined as open to air to ensure only disc rotation influenced the flow. A 3D structured hexahedron mesh was constructed in ANSYS Design Modeler (Fig. 2b) based on the geometry described above. Several mesh densities were tested to confirm grid independence.

The mesh size was varied from 200,000 to 700,000 elements based on mesh quality metrics, including skewness, aspect ratio and orthogonal quality, the wall y^+ value, and bias factors used for radial mesh refinement, as shown in detail in the supporting information (Fig. S1). The wall y^+ is a dimensionless parameter that characterizes near-wall mesh resolution by relating the friction velocity to fluid properties, thereby indicating whether the wall-adjacent cells are located within the viscous sublayer or the turbulent near-wall region [38]. The skewness parameter measures the deviation of a cell from an ideal equilateral or equiangular geometry. The aspect ratio represents the ratio between the longest and shortest dimensions of a computational cell. The orthogonal quality evaluates the alignment between the face normal vectors and the vectors connecting adjacent cell centroids. The recommended ranges of these mesh quality metrics are summarized in Table S2 while the distributions of their localized values are shown in Fig. S2. In addition, the key hydrodynamic parameters used for grid independence verification, including liquid hold-up over the rotating disc, average liquid film thickness and averaged turbulent kinetic energy (TKE), are presented in Tables S3, S4, and S5 for the smooth disc, P36 disc, and P24 disc, respectively.

A refined boundary layer near the disc surface was created with a varying y^+ value of 1.68 to 0.05 and a suitable radial bias factor (see the details in the supporting information, Table S1), ensuring that the smallest cell dimension remained below the critical feature size. The y^+ selection is consistent with previous CFD modeling studies in SDR which suggested a y^+ of approximately 0.2 [39]. The highest mesh



(a) 3D geometrical model



(b) 3D mesh model

Fig. 2. Schematic drawings of SDR models.

Table 1

Physical properties of refined palm oil and methanol.

Properties	Refined palm oil	Methanol
Density (kg/m ³)	886	755
Dynamic viscosity (Pa.s)	0.01946	0.00036
MW (g/mol)	870.2	32.04

density, containing 559,500 hexahedral cells and providing a wall y^+ value of 0.14, was selected for the smooth disc surface to accurately resolve the flow field. The corresponding average skewness, average aspect ratio, and orthogonal quality values were 0.52, 37.57, and 0.99, respectively. The high skewness values were confined mainly to the inlet region due to domain partitioning. In contrast, the maximum aspect ratio of 224.93 occurred near the rotating disc surface, where highly stretched cells were used to resolve the thin liquid film and steep velocity gradients normal to the wall. The cells associated with high skewness accounted for <1% of the total mesh volume and are therefore expected to have limited influence on the overall numerical results. For the roughened discs, a mesh consisting of 669,720 hexahedral cells was selected, providing a wall y^+ value of 0.05. In addition, the variations in the key hydrodynamic parameters between each mesh and finest mesh resolutions were below 5% in all indicators for smooth disc (Table S4), whereas the deviations in TKE and liquid volume hold-up are

approximately 7% for both the P24 and P36 cases (Tables S5 and S6).

Grid independence was further verified using the average liquid film thickness profile along the disc radius for the smooth, P36, and P24 discs, as shown in Fig. 3. Although the 445,000-cell mesh exhibited good agreement with the theoretical values obtained from the Nusselt model, employing the finer mesh (559,500 cells) can improve the resolution of the liquid-liquid interface between palm oil and methanol, which is critical for analyzing mixing and interfacial behavior, as shown in Fig. 4.

2.3.2. Interface modeling

The flow in the SDR is treated as a thin film of two immiscible liquids, refined palm oil and methanol, and is modeled using the Volume of Fluid (VOF) method. In this formulation, a scalar transport equation for the phase fraction of one fluid is solved in conjunction with the incompressible continuity and Navier–Stokes momentum equations as depicted in Eqs. (5) and (6), respectively. Under the assumptions of isothermal, incompressible flow, the continuity equation enforces mass conservation, while the momentum equations account for pressure, viscous, and interfacial forces. The VOF phase-fraction field then reconstructs the instantaneous interface and determines local mixture properties throughout the domain as presented in Eqs. (7)–(9).

Continuity equation:

$$\frac{\partial \alpha_q}{\partial t} + \nabla \cdot (\alpha_q \mathbf{u}) = 0 \quad q = g \text{ or } l \quad (5)$$

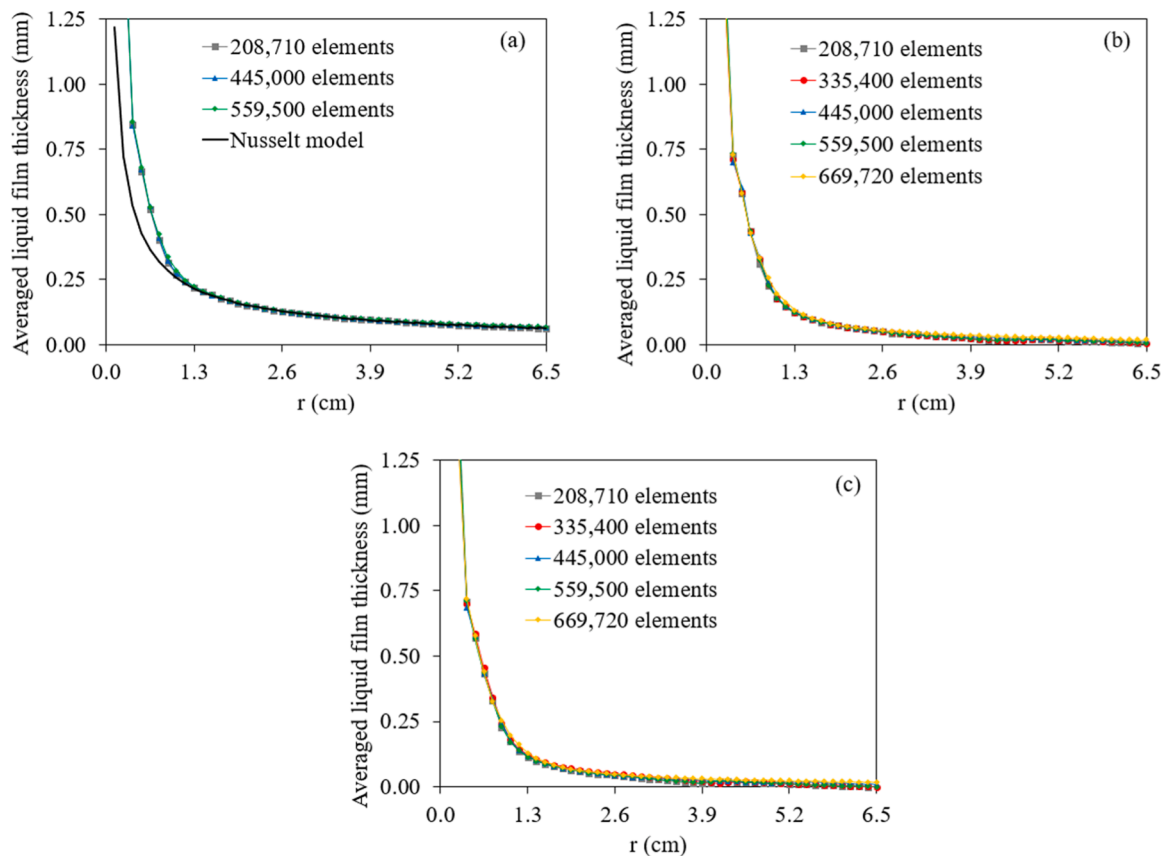


Fig. 3. Grid independence analysis via average liquid film thickness from the Nusselt model at a rotational disc speed (R) of 1500 rpm for (a) smooth disc surface, (b) P36 disc surface ($\approx 538 \mu\text{m}$ average roughness) and (c) P24 disc surface ($\approx 764 \mu\text{m}$ average roughness).

Momentum equation:

$$\frac{\partial (\rho u)}{\partial t} + \nabla \cdot (\rho u u) = -\nabla p + \nabla \cdot [\mu (\nabla u + \nabla u^T)] + \rho g + F_\sigma \quad (6)$$

where u is the mean velocity vector of the fluids (m/s), p is the mean static pressure (Pa), F_σ is the surface tension force (N/m³). The phase-fraction fields in the VOF model are coupled by the requirement that all local volume fractions sum to unity:

$$\sum_{q=1}^n \alpha_q = 1 \quad (7)$$

where α_q is a volume fraction of the q phase. The physical parameters of mixed phases such as ρ and μ can be obtained by the weighted average of the volume fraction of each component:

$$\rho = \sum_{q=1}^N \alpha_q \rho_q \quad (8)$$

$$\mu = \sum_{q=1}^N \alpha_q \mu_q \quad (9)$$

where ρ and μ denote the density and dynamic viscosity of phase q , respectively.

2.3.3. Turbulence modelling

The SST k - ω model is a widely adopted two-equation turbulence formulation that simultaneously solves transport equations for turbulent kinetic energy (k) and its specific dissipation rate (ω). By employing a blending function (F_1), it transitions from a k - ω formulation in the near-wall viscous sublayer to a k - ϵ formulation in the outer flow, thereby

ensuring both high-fidelity boundary-layer resolution and robust performance at elevated Reynolds numbers (Re) [40]. Given the coexistence of intense shear adjacent to the disc surface and turbulent behavior in the bulk film, this hybrid approach is particularly suited to SDR thin-film flows. The k and ω parameters for the SST k - ω turbulence model are stated in the transport equations as shown in our previous work [32].

2.3.4. Boundary conditions and calculation procedure

Simulations were carried out using ANSYS Fluent 2023 R2 in double-precision mode. The CFD simulations were performed under an isothermal condition at 60 °C, corresponding to the measured reactor-inlet temperature of both liquid feeds and the reactor operating temperature. Since reaction kinetics and species transport were not included, the CFD results were interpreted as hydrodynamic and mixing-related descriptors rather than direct predictions of FAME yield. In the experiments, NaOH was dissolved in methanol before feeding. In the CFD model, however, NaOH was not explicitly included as a transported species, and the methanol/NaOH feed was approximated using methanol phase properties. This simplifying assumption is consistent with the non-reactive hydrodynamic scope of the CFD model, which was used to compare film thickness, phase distribution, residence time, velocity field, and turbulence-related quantities rather than species concentration or FAME yield. The rotating disc surface was defined as a no-slip rotating wall with angular velocity matched to each experimental rotational speed, while all other solid boundaries were treated as stationary no-slip walls. Remaining domain faces were set as open to ambient air to isolate disc-rotation effects. Surface roughness was incorporated via the standard wall-roughness model, with roughness heights of 538 μm (P36) and 764 μm (P24) assigned to the corresponding disc variants [41].

Flow regime selection (laminar or SST k - ω turbulence) was governed

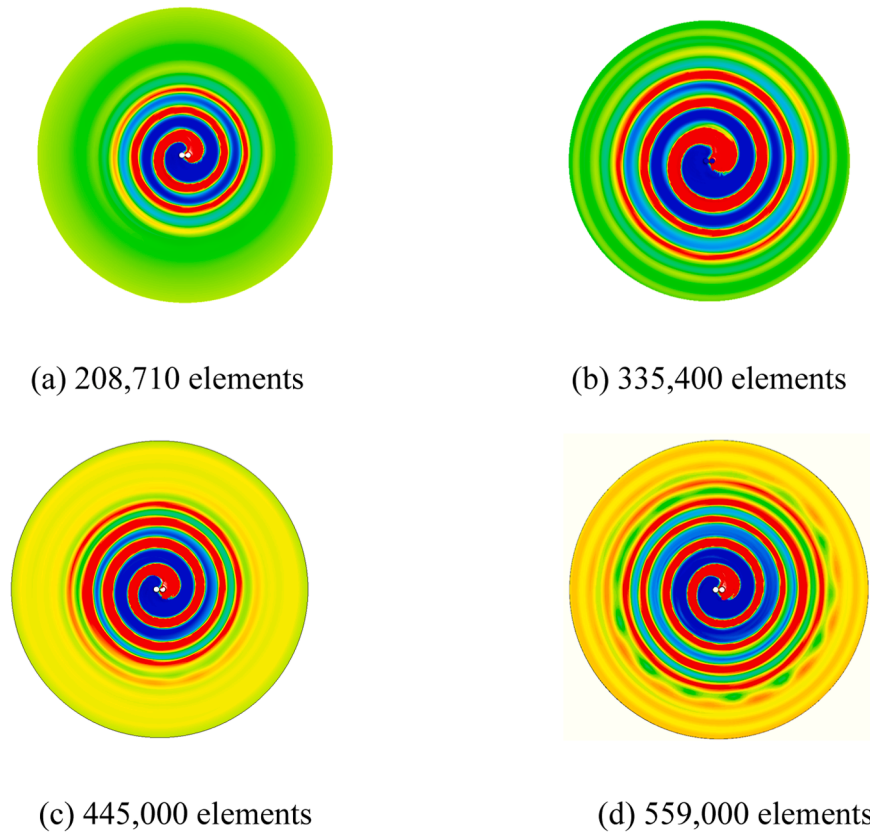


Fig. 4. Grid independence analysis via methanol volume fraction contour, at R of 1500 rpm for a smooth disc surface as an indicator.

by the combination Reynolds number (Re) as detailed in our previous work [32]. The SST $k-\omega$ model was selected due to its capability to predict the overall hydrodynamic behavior of thin-film flow in spinning-disc reactors with reasonable computational cost. However, because the SST $k-\omega$ model is based on the Reynolds-averaged Navier–Stokes (RANS) framework, it cannot directly resolve transient microscale turbulent structures or Kolmogorov-scale eddies. Therefore, the predicted turbulence characteristics should be interpreted as time-averaged flow behavior rather than directly resolved small-scale turbulent structures. Pressure-velocity coupling was handled by the SIMPLE algorithm [42], and convective terms were discretized using the QUICK scheme [43]. A residual convergence criterion of 10^{-3} was enforced for all equations. An adaptive time step size was employed to ensure numerical stability by controlling the Courant-Friedrichs-Lewy (CFL) number [44]. The maximum CFL number was limited to 0.2 based on previous studies of rotating flow systems [45].

2.3.5. Model validation

Validation of the CFD model was performed by comparing simulated liquid film thicknesses against the regression-based prediction models of Wang et al., which are based on a modified Nusselt correlation [46]. Specifically, the regression model (Eq. (9)) and its underlying Nusselt form (Eqs. (10) and 11) were applied, with the non-dimensional radial coordinate ($\hat{r}$), defined in Eq. (12).

$$h = 1.06 h_N \left(\frac{Q^2}{r^4 \nu \Omega} \right)^{\frac{1}{48}} \quad (10)$$

$$h_N = \left(\frac{3Q\nu}{2\pi r^2 \Omega^2} \right)^{\frac{1}{3}} \quad (11)$$

$$\hat{r} = \frac{r}{\left(\frac{Q^2}{\nu \Omega^2} \right)^{0.25}} \quad (12)$$

The Nusselt model for liquid film thickness prediction is fundamentally derived based on the assumption of fully developed laminar thin-film flow over a rotating disc. Consequently, the predicted film thickness is generally lower than the actual film thickness because the model does not account for surface waviness and turbulent fluctuations within the liquid film. Nevertheless, previous studies reported that the deviation between experimental measurements and Nusselt-based predictions is generally within approximately 10% under centrifugal force-dominated flow conditions, particularly when the Ekman number (E_k), expressed in Eq. (13), is greater than 10 [47,48].

$$E_k = \left(\frac{2\pi}{3} \right)^{\frac{2}{3}} \left(\frac{r^4 \Omega \nu}{Q^2} \right) \quad (13)$$

2.3.6. Mean residence time calculation

The mean residence time over the rotating disc was calculated from the total liquid hold-up volume beneath the liquid film (V_{film}) divided by the volumetric flow rate (Q) as shown in Eq. (14).

$$\text{Mean residence time } (\tau) = \frac{V_{film}}{Q} \quad (14)$$

The liquid hold-up volume was obtained through CFD post-processing based on the liquid volume fraction distribution beneath the air-liquid interface. Since the predicted liquid film thickness showed a deviation below 10%, the estimated liquid hold-up volume and mean residence time are considered to provide a reasonable representation of the overall flow behavior in the present study.

3. Results and discussion

3.1. Model validation

Model validation is essential in CFD simulation to ensure the reliability of the predicted hydrodynamic behavior within the spinning disc reactor (SDR). However, direct experimental measurement of the liquid film formed on the spinning disc, particularly at high rotational speeds, remains challenging. Consequently, the prediction liquid film model, as the modified Nusselt model, is commonly adopted as a benchmark reference for validating the CFD-SDR simulations [32,33,46,49].

Previous studies have investigated the behavior of liquid film thickness in the SDR using both experimental measurements and CFD simulations to develop the Nusselt prediction model [32,33,46,49]. Experimental liquid film thickness obtained using laser-based high-speed imaging techniques, while CFD simulations employing the VOF method coupled with a laminar viscous model were used for comparison. These comparisons supported the use of Nusselt-based models as a practical benchmark for overall film-thickness trends, although they do not constitute full validation of interfacial mixing or turbulence quantities. In addition, the experimental liquid-film-thickness data reported by Burns et al. [50] obtained using an electrical resistance technique, were used as benchmark data in subsequent CFD-SDR studies. For example, VOF-based CFD simulations, including studies using the SST $k-\omega$ turbulence model, were employed to predict and compare liquid-film thickness with the experimental data of Burns et al. The results demonstrated reasonable agreement between the experimental and CFD data, while the deviation between CFD predictions and the Nusselt model was reported to be within approximately 15%. Accordingly, the modified Nusselt model remains a practical and widely accepted approach for validating CFD-SDR simulations, particularly under centrifugal force dominated conditions where the Ekman number (Ek) > 10.

Fig. 5 shows the CFD model with a smooth disc matches the modified

Nusselt correlation of Wang et al. [46] across disc speeds of 500, 1000 and 1500 rpm at a total flowrate of 270 mL/min. Two characteristic radii appear in every profile, the injection radius (r_i) and the synchronization radius (r_s). Between these points, the flow on the spinning disc can be divided into three distinct hydrodynamic zones. (i) The injection zone ($r < r_i$), the film thickness is controlled by the impact momentum of the refined palm oil and methanol streams hitting the disc surface. In the injection zone ($r < r_i$), the CFD-predicted film thickness was slightly higher than that predicted by the Nusselt-based correlation near the inlet. This deviation is attributed to inlet impingement effects, which are not included in the correlation model. (ii) Acceleration zone ($r_i < r < r_s$), as the liquid moves radially outward, centrifugal forces intensify, causing a rapid reduction in film thickness. Across all rotational speeds, simulation and correlation differ by no more than about 8% at any rotational speeds. (iii) Synchronization zone ($r > r_s$), the near-wall tangential liquid velocity approaches the local tangential velocity of the rotating disc due to the no-slip boundary condition. This angular momentum transfer enhances centrifugal radial transport, resulting in a minimum liquid-film thickness. Reasonable agreement was observed between CFD predictions and the Nusselt correlation for $r' > 0.9$, indicating that the model can reasonably reproduce overall thin-film hydrodynamic trends. The discrepancy between the simulation and Nusselt correlation was lower than 5% for the three rotational speeds of 500, 1000, and 1500 rpm at $r = 0.0561$ m for the smooth disc. Therefore, the present validation mainly focuses on evaluating the overall liquid film behavior and hydrodynamic trends rather than capturing detailed localized turbulent interfacial structures. Although the Nusselt-based comparison provides a useful benchmark for liquid-film thickness, it does not fully validate interfacial mixing, TKE distribution, or local phase-contact behavior. Therefore, CFD-derived quantities such as phase distribution, residence time, and TKE are interpreted as comparative hydrodynamic descriptors. More rigorous validation would require additional experimental measurements, such as RTD, droplet-size

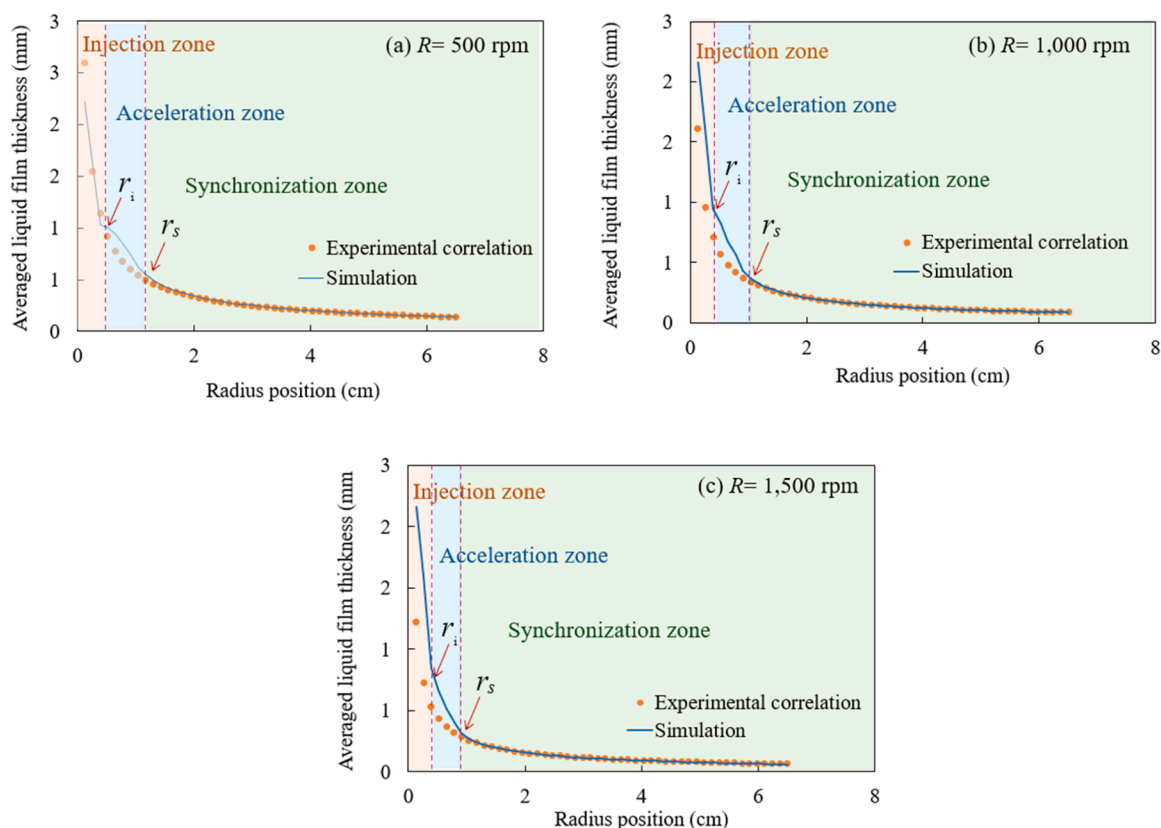


Fig. 5. Comparison of liquid film thickness between simulation and Nusselt correlation results in a smooth disc ($Q = 270$ mL/min, $R = 500, 1000$ and 1500 rpm).

distribution, local concentration profiles, or conversion profiles.

3.2. Hydrodynamic insight from CFD

Fig. 6 shows the CFD-predicted methanol volume fraction contours for the refined palm oil–methanol system on the smooth, P36 and P24 discs at rotational speeds of 500, 1000, and 1500 rpm. The contours were used to compare phase-distribution patterns among the disc surfaces and rotational speeds. For the smooth disc, methanol-rich regions formed spiral patterns along the liquid film. As the rotational speed increased, the spiral bands became narrower and more elongated, suggesting faster radial transport and stronger shear-induced redistribution of the methanol phase. Disc-surface roughness altered the phase-distribution pattern within the liquid film. Compared with the smooth disc, the P36 and P24 discs exhibited more persistent methanol-rich spiral structures over the disc surface, suggesting stronger near-wall disturbance and improved oil–methanol phase contact under the tested conditions. This behavior is consistent with previous SDR studies showing that roughened, grooved, or modified disc surfaces can increase wall friction, induce surface-driven perturbations, and promote micromixing-related hydrodynamic behavior [32,51,52]. For the P24 disc, the phase distribution became more dispersed, particularly at higher rotational speeds, suggesting a stronger influence of surface roughness on methanol-phase distribution and interfacial contact. However, these volume-fraction contours should be interpreted qualitatively as hydrodynamic indicators of phase distribution rather than direct measurements of mixing efficiency or mass-transfer coefficient.

To quantify the thin-film hydrodynamics from Fig. 6, the radial distribution of liquid-film thickness was extracted from CFD results. Fig. 7 presents the time-averaged film thickness versus radial position at 1000 rpm. For all disc surfaces, the film thins rapidly near the inlet and approaches a stable value toward the outer radius, driven by centrifugal force and radial transport. Notably, the structured discs (P36 and P24) consistently produce thinner films than the smooth disc. In the outer

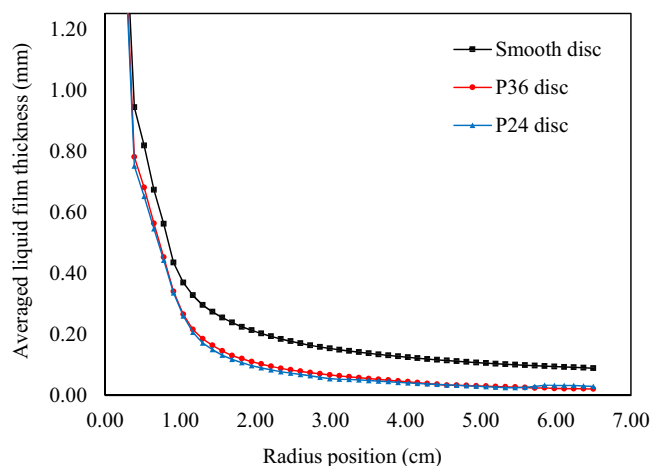


Fig. 7. Radial profiles of averaged liquid-film thickness at $R = 1000$ rpm for the smooth, P36 and P24 disc surfaces (CFD results).

region ($r \approx 4\text{--}6.5$ cm), the smooth disc maintains a thickness of 0.089–0.124 mm, whereas P36 and P24 yield significantly lower values of 0.021–0.044 mm. The P24 disc produced the thinnest film overall, which is consistent with stronger momentum exchange and enhanced surface renewal, supporting the more dispersed phase-distribution pattern observed in Fig. 6. These thickness trends provide a possible hydrodynamic link between disc-surface modification and reactor performance. Thinner films can increase velocity gradients and may promote oil–methanol phase redistribution and interfacial contact within the liquid film.

Furthermore, the influence of rotational speed and disc surface roughness on the turbulent kinetic energy (TKE) was investigated via CFD simulation to demonstrate the enhancement of turbulence in the SDR. TKE is a key parameter defined as the intensity of turbulence

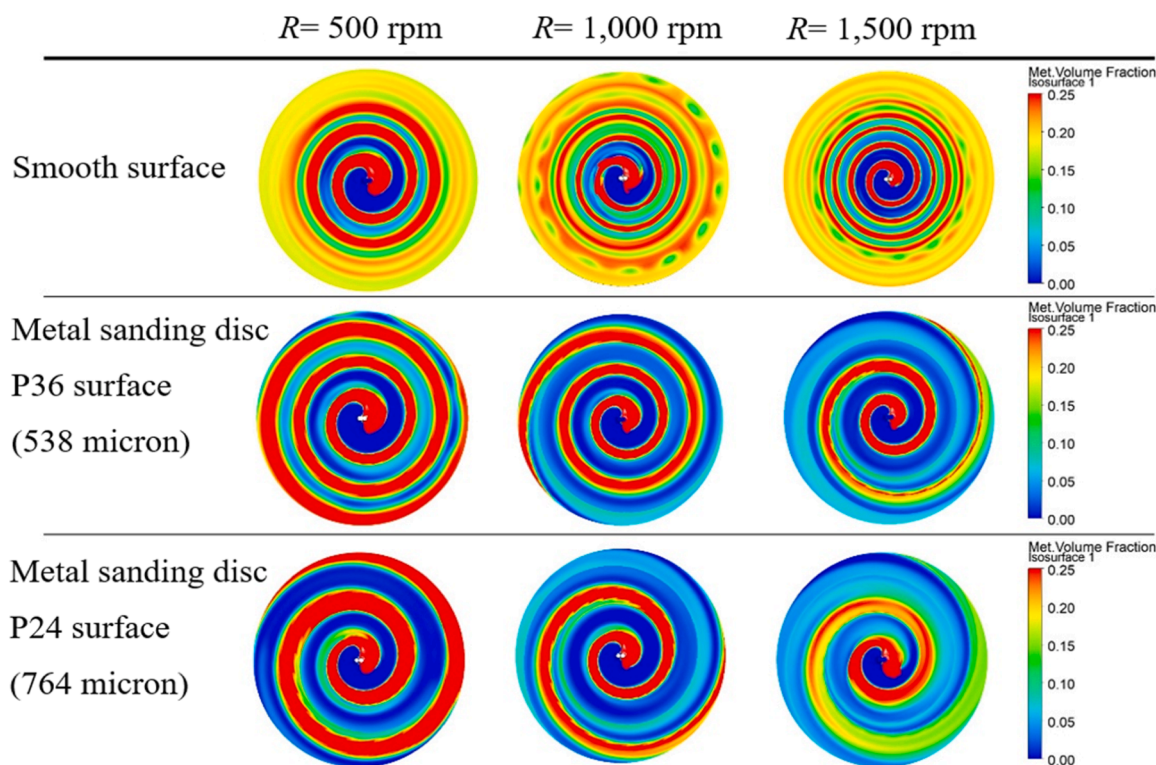


Fig. 6. Methanol volume fraction contours of the refined palm oil–methanol flow on different surfaces (smooth surface and rough surfaces of P36 and P24) at various rotational disc speeds (500, 1000 and 1500 rpm).

within a flow field. Large TKE leads to the increment of the turbulent dissipation rate energy, which promotes the breakage of larger eddies into smaller (Kolmogorov scale) eddies [53]. In the context of a SDR, the energy produced from the rotating disc is transferred from the disc surface to the liquid film, where it promotes interfacial mixing between immiscible phases [32,33]. This transfer enhances the overall mixing efficiency by extending the synchronization zone (i.e., shifting r_s inward). Accordingly, the enlarged synchronization zone may promote the formation of characteristic interfacial wave structures, known as spiral pulse-like waves (SPW) and spiral sine-like waves (SSW), which are associated with intensified turbulent behavior beneath the liquid film [30,54]. Because the present RANS-based model cannot directly resolve microscale eddies, these wave structures are interpreted as indicators of enhanced interfacial disturbance and turbulence-related activity rather than as directly resolved Kolmogorov-scale structures. The increased TKE therefore reflects modeled turbulent energy transfer and possible eddy-breakdown tendency within the resolution limits of the model.

In previous CFD-SDR study, SDR was applied in photochemical reactions employed CFD simulations to investigate hydrodynamic behavior within the liquid film, while the reaction performance was evaluated based on photon absorption rate [33]. The rotating disc surface was modified by attaching spoilers onto the disc surface to intensify mixing. The simulation results showed that fluid flowing through the spoiler structures generated vortices downstream of the middle and outer spoiler layers, corresponding to regions of high TKE intensity observed in the TKE contours. In addition, the mean radial TKE profile increased progressively along the disc radius and reached higher magnitudes for the spoiler-modified disc. The increased TKE was associated with significantly improved photon absorption performance, suggesting enhanced mixing and interfacial transport behavior within the SDR. Therefore, TKE was used as a qualitative hydrodynamic descriptor for comparing turbulence enhancement among disc surfaces.

The color-coded maps of TKE contours at 1000 rpm are presented in Fig. 8 at different distances from the disc surface ($z = 50, 150, 250$, and $500 \mu\text{m}$). A distinct red contour, indicating the zone of higher TKE, was observed and its trend increased with both rotational speed and roughness of disc surface. The higher TKE zone was in agreement with the progressively smoother red contours in the refined palm oil-methanol flow pattern, as mentioned in Fig. 6, as the improved

centrifugal force. For the smooth disc, relatively low TKE values were observed over most of the liquid-film region, although localized spiral-like high-TKE regions appeared at larger distances from the disc surface. For both P24 and P36 roughened discs, higher TKE values were obtained near the outer radius, appearing as red-colored regions. These regions indicate enhanced modeled turbulence intensity and may correspond to stronger interfacial disturbances, including characteristic wave structures such as *spand* and SSW. However, these features should not be interpreted as directly resolved microscale eddies because the present RANS-based model provides time-averaged turbulence quantities. Therefore, the increase in TKE should be interpreted as a qualitative hydrodynamic indication of enhanced modeled turbulence intensity rather than direct evidence of resolved eddy formation or mixing efficiency. Among the tested discs, P24 showed the strongest turbulence-related response, suggesting that increased surface roughness may promote local agitation and interfacial renewal under the tested conditions.

3.3. Correlation between hydrodynamics and biodiesel yield

It should be noted that the CFD model did not include reaction kinetics or species transport. Therefore, CFD-derived quantities such as film thickness, phase distribution, residence time, velocity field, and TKE are interpreted as hydrodynamic descriptors associated with the experimental FAME yield trends, rather than as direct predictors of FAME yield. In addition, previous studies on conventional biodiesel via transesterification have reported that oil-methanol immiscibility can lead to mass-transfer-controlled behavior during the initial stage of reaction and that mixing intensity can influence reaction progress [55,56]. Based on this background, the present study was designed to examine how hydrodynamic modification in the SDR affects the experimentally observed FAME yield under fixed chemical conditions. The reaction temperature, methanol-to-oil molar ratio, NaOH loading, feed composition, and total feed flow rate were kept constant, while disc-surface texture and rotational speed were varied. Therefore, the FAME yield trends are discussed in relation to thin-film hydrodynamics, phase distribution, residence time, and turbulence-related behavior under the tested conditions.

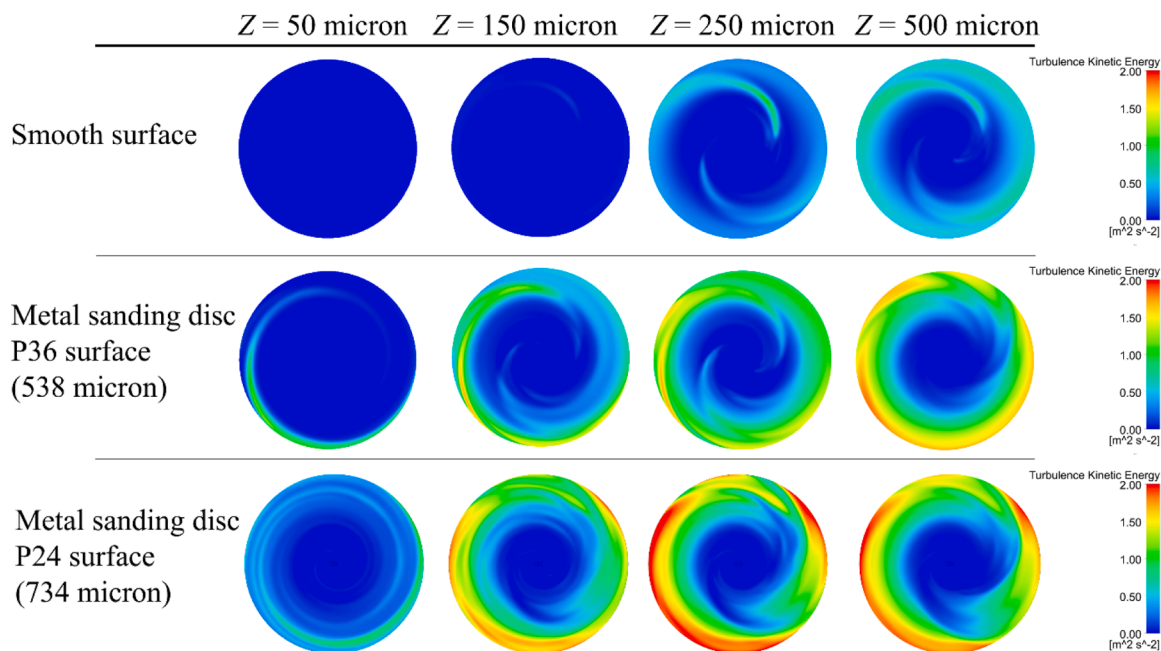


Fig. 8. Turbulence kinetic energy (TKE) for smooth surface and rough surfaces (P36 and P24) at a rotational disc speed of 1000 rpm.

3.3.1. Effect of disc-surface roughness

The effect of disc-surface roughness on biodiesel production performance was evaluated by comparing the smooth, P36 and P24 disc surfaces at the same operating conditions. As shown in Fig. 9, disc-surface roughness markedly affected FAME yield across the entire range of rotational speeds investigated. At 500 rpm, the FAME yield increased from 70.0% for the smooth disc to 78.2 and 82.1% for the P36 and P24 discs, respectively. A similar trend was observed at 1000 rpm, where the yield increased from 84.6% (smooth) to 90.0% (P36) and further to 97.3% (P24). At 1500 rpm, the roughened discs consistently outperformed the smooth disc, yielding 88.8% for P36 and 96.9% for P24, compared with 82.3% for the smooth surface. These results indicated that disc-surface roughness is associated with improved FAME yield under fixed feed composition, reaction temperature, and total feed flow rate. The experimentally observed performance order (P24 > P36 > smooth) was consistent at all rotational speeds, suggesting that surface texturing can act as a passive hydrodynamic intensification strategy in the SDR.

The yield enhancement with roughness was consistent with CFD-predicted changes in near-wall hydrodynamics and phase distribution. As can be seen from the methanol volume fraction contours (Fig. 6), roughened surfaces (P36 and P24) exhibited more persistent methanol-rich spiral structures compared with that of the smooth disc across the studied speeds, indicating sustained interfacial contact between methanol and refined palm oil within the thin liquid film. This behavior is consistent with previous SDR studies showing that roughened, grooved, or modified disc surfaces can increase wall friction, induce surface-driven perturbations, and promote micromixing-related hydrodynamic behavior [32,51,52]. In addition, the TKE contours at 1000 rpm (Fig. 8) show higher dissipation levels for the roughened discs, with the most pronounced increase observed for P24. The average TKE values at 1000 rpm were 1.52×10^{-4} , 5.97×10^{-3} and 3.15×10^{-2} J/kg for the smooth disc, P36 and P24, respectively (P24 > P36 > smooth). These results suggest that increased surface roughness enhances modeled turbulence-related hydrodynamic activity and may promote stronger interfacial disturbance within the liquid film [53]. However, because the present RANS-based model provides time-averaged turbulence quantities, the TKE results should be interpreted as qualitative hydrodynamic indicators rather than direct evidence of resolved small-scale eddy formation or mixing efficiency.

3.3.2. Effect of rotational disc speed

Rotational disc speed strongly influenced the steady-state FAME yield for all disc surfaces, but the dependence was non-monotonic

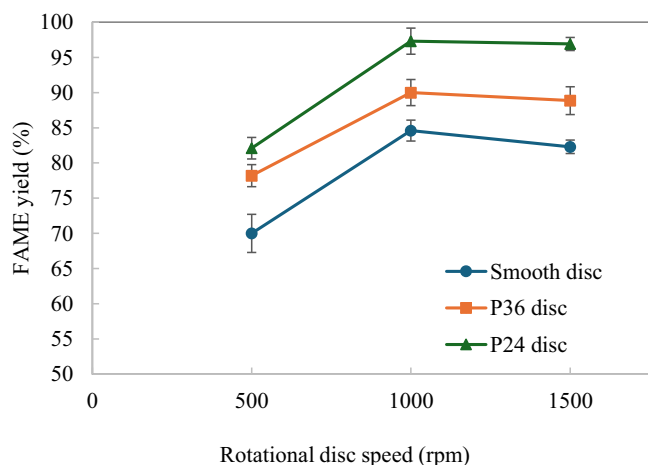


Fig. 9. FAME yield as a function of rotational disc speed for smooth, P36 and P24 discs (feed flow rate = 270 mL/min, methanol-to-oil molar ratio = 6, NaOH loading = 0.75 wt.% of oil, and $T = 60^\circ\text{C}$).

(Fig. 9). Increasing the speed from 500 to 1000 rpm resulted in substantial yield improvements: from 70.0 to 84.6% for the smooth disc, from 78.2 to 90.0% for P36, and from 82.1 to 97.3% for P24. However, further increasing the speed from 1000 to 1500 rpm led to a decrease or plateau in yield, from 84.6 to 82.3% for the smooth disc, from 90.0 to 88.8% for P36, and a slight reduction from 97.3 to 96.9% for P24. Thus, among the three rotational speeds tested, 1000 rpm provided the highest FAME yield under the present operating conditions. Beyond which additional increases in speed did not facilitate a higher steady-state FAME yield. This could be supported by CFD-predicted flow patterns and interpreted as a balance between interfacial formation/mixing enhancement and the time available for reaction within the thin film. From 500 to 1000 rpm, the methanol volume fraction contours (Fig. 6) show enhanced spiral development and narrower spiral bands on the smooth disc, suggesting improved phase redistribution and enhanced oil-methanol interfacial contact within the thin liquid film. This hydrodynamic behavior is consistent with the increase in FAME yield observed for all disc surfaces over the same rotational-speed range. This interpretation was further supported by the CFD-derived averaged turbulent kinetic energy (TKE, Fig. 10a), which markedly increased on the structured discs as the speed increased (e.g., at 1000 rpm, TKE rises to 5.97×10^{-3} J/kg for P36 and 3.15×10^{-2} J/kg for P24, while remaining much lower for the smooth disc), indicating stronger velocity fluctuations and interfacial renewal. At very high rotational speed (1000 to

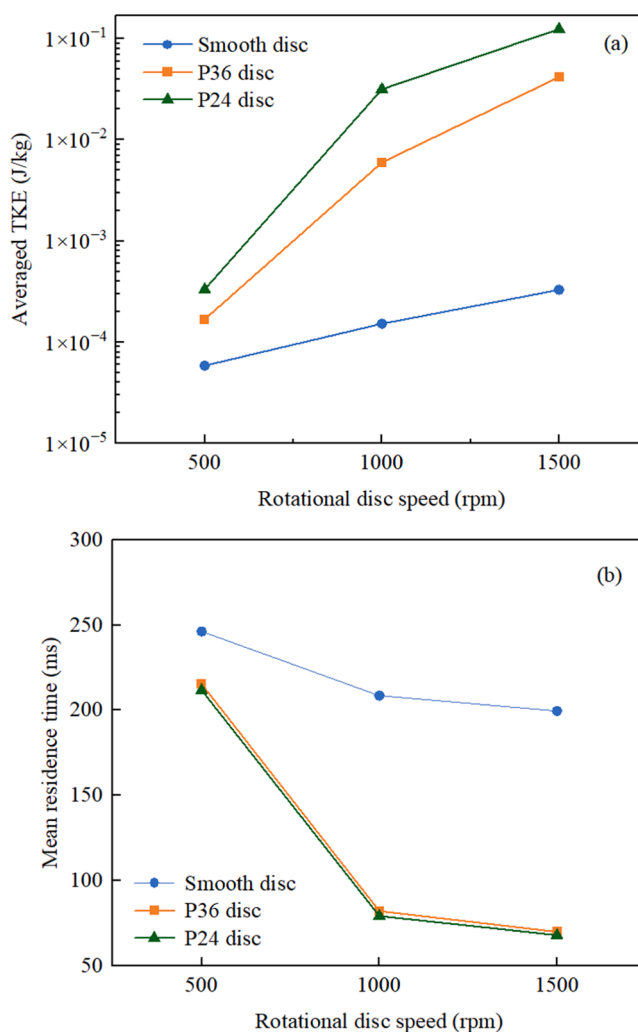


Fig. 10. Averaged turbulence kinetic energy (TKE) plotted on a logarithmic y-axis (a) and Mean residence time of reactants (b) as a function of rotational speed for smooth, P36, and P24 discs ($Q = 270$ mL/min).

1500 rpm), although the spiral structures remain strongly developed, the effective residence time and film hold-up available for reaction can become limited due to faster radial transport. The calculated residence time on the rotating disc was decreased with increasing speed as presented in Fig. 10b. At 1500 rpm, the mean residence time decreases to 199.5 ms for the smooth disc (from 208.6 ms at 1000 rpm), and to 70.0 ms and 68.1 ms for P36 and P24, respectively. This residence time reduction may limit the contact time available for transesterification, even when hydrodynamic agitation is enhanced. For the P24 disc, the high TKE value at 1500 rpm (1.2×10^{-1} J/kg) suggests a stronger turbulence-related hydrodynamic response than that of P36 (4.2×10^{-2} J/kg). This response was consistent with more dispersed spiral structures and stronger interfacial disturbance, as reported in previous SDR hydrodynamic studies [32,57]. This hydrodynamic behavior may partially offset the reduced residence time and are consistent with the nearly unchanged FAME yield of the P24 disc at the highest rotational speed.

3.4. Performance comparison with different intensified reactors

The performance comparison in this section is intended as contextual benchmarking rather than a direct ranking of reactor technologies. This is because the reported systems differ in feedstock type, catalyst, methanol-to-oil molar ratio, reaction temperature, residence time, flow rate, reactor scale, and energy-accounting boundary. In the present work, the yield efficiency was calculated using Eq. (2) based on the defined experimental system boundary. The measured energy consumption included the spinning disc reactor, peristaltic pumps, feed-line heating tape, and water baths used for feed thermal conditioning and reactor temperature control. In addition to yield efficiency, FAME productivity was included as an additional performance indicator because it accounts for the FAME production rate relative to the reactor volume.

The performance of the roughened P24 SDR was compared with other intensified reactors to evaluate its potential for continuous biodiesel production. As shown in Table 2, the P24 SDR reached a FAME yield of 97.3% in a disc residence time of <1 s under a reaction temperature of 60 °C and a methanol-to-oil molar ratio of 6:1. This result suggested that the textured SDR can mitigate oil-methanol mass transfer resistance under the tested conditions, as the surface roughness may disturb the thin liquid film and generate localized mixing zones near the disc surface. However, this result could not be interpreted as complete elimination of mass transfer limitations, since the present CFD model does not include reaction kinetics or direct mass-transfer-coefficient

prediction.

Compared with the smooth disc SDR, the P24 disc increased the yield efficiency from 1.370×10^{-3} to 1.402×10^{-3} g/J. This improvement indicates that disc-surface texturing enhanced FAME yield efficiency under the defined experimental boundary. The rotating tube reactor offered a higher yield efficiency (3.750×10^{-3} g/J); however, it required a residence time of 11.2 min, which is substantially longer than the short disc residence time used in the SDR. When the FAME productivity (P_{FAME}/V_R) was considered, the RTR showed a lower value than the P24 SDR. This difference reflects the different hydrodynamic mechanisms of the reactors. The rotating tube reactor relies mainly on vortex-driven mixing, whereas the SDR relies on rapid thin film spreading and high shear at the rotating surface. The microwave-assisted system gave a high FAME yield but used much more energy, leading to a lower yield efficiency. In the case of non-catalytic glycerolysis, the microwave reactor required a high methanol-to-oil molar ratio of 30:1 to achieve a higher yield efficiency. This high excess of methanol adds significant costs and energy for methanol recovery after the reaction. The microchannel reactor also gave a high yield but at a very low flow rate of 0.2 mL/min, which is not practical for large-scale production compared to the 270 mL/min used in this work. Similarly, the multiple baffle reactor needed 10 min to reach a 97.7% FAME yield. Overall, the P24 SDR showed competitive performance under the investigated laboratory-scale conditions and may serve as an alternative intensified reactor for continuous biodiesel production. Nevertheless, industrial-scale application would require further evaluation of scale-up behavior, film stability at higher throughput, system energy demand, equipment complexity, and long-term durability of the textured disc surface. For industrial-scale application, the optimum flow rate and rotational speed identified in this laboratory-scale SDR could not be directly transferred to larger systems. SDR scale-up requires hydrodynamic similarity to maintain comparable film thickness, flow regime, residence time, and mixing intensity. Dimensionless numbers such as Reynolds, Weber, and Ekman numbers can be used to guide scale-up, while disc enlargement and numbering-up may be combined to achieve higher throughput. Therefore, further scale-up studies are required before industrial implementation [23,58].

4. Conclusions

This study investigated continuous biodiesel production in a spinning disc reactor (SDR) and examined the effects of disc rotational speed and surface roughness on thin-film hydrodynamics, mixing-related

Table 2
Performance comparison with other intensified reactors for biodiesel production.

Reactor	Feedstock	Catalyst	C _{cat} (wt. %)	MeOH to oil molar ratio	T (°C)	Flow rate (mL/ min)	Residence time (min)	FAME yield (%)	Yield efficiency $\times 10^3$ (g/J)	FAME productivity $P_{\text{FAME}}/V_R \times 10^3$ (mol/ (mL·min))	Key mechanism	Ref.
SDR (Roughness P24 surface)	RPO	NaOH	0.75	6	60	270	79.3 ms	97.3	1.402	50.0	Thin film + high shear + roughness-induced interfacial disturbance	This work
SDR (Smooth disc)	RPO	NaOH	1	6	60	260	<2 s	97.0	1.370	48.0	Thin film + high shear	[29]
Rotating tube reactor (RTR)	RPO	NaOH	1	6	30	30	11.2	97.5	3.750	0.214	Taylor-vortex-driven mixing	[14]
Microchannel reactor	RPO	KOH	5	7.6	25	0.2	40 s	98.6	N/A	3.585	High interfacial area / Short diffusion length	[59]
Continuous-flow microwave reactor	RPO	NaOH	1	12	70	100	1.75	99.4	0.287	1.143	Volumetric heating / fast heating	[60]
Zigzag micro-channel reactor	Soybean oil	NaOH	1	5.5	60	0.27	18 s	97.3	N/A	7.667	High interfacial area / Short residence time	[8]

RPO = refined palm oil.

WCO = waste cooking oil.

behavior, and FAME yield. A CFD framework based on the VOF approach coupled with the SST $k-\omega$ turbulence model was developed and validated using a modified Nusselt correlation for thin-film thickness, confirming that the model can reliably reproduce thin-film hydrodynamics under the investigated operating conditions. Experimentally, disc-surface texturing was associated with improved FAME yield under the investigated conditions. At the three rotational speeds tested, the roughened discs gave higher FAME yields than the smooth disc, and the performance order $P24 > P36 > \text{smooth}$ was consistently observed. At 1000 rpm, the highest FAME yield among the tested conditions, 97.3%, was obtained using the P24 disc, compared with 90.0% for P36 and 84.6% for the smooth disc. Among the three rotational speeds investigated, 1000 rpm provided the highest FAME yield. Although increasing speed from 500 to 1000 rpm markedly improved yield for all disc surfaces, a further increase to 1500 rpm resulted in a decrease in yield, indicating a trade-off between hydrodynamic mixing enhancement and the effective residence time available for reaction in the thin film.

CFD-derived methanol volume fraction contours indicated that roughened surfaces promoted more persistent methanol-rich spiral structures, suggesting enhanced phase redistribution and oil-methanol interfacial contact within the thin liquid film. The P24 surface showed the strongest turbulence-related hydrodynamic response, as indicated by the CFD-predicted TKE distribution. Finally, energy-normalized benchmarking against representative intensified reactor configurations showed that the present SDR with a P24 surface achieved high FAME yield, a yield efficiency of 1.402×10^{-3} g/J and high volumetric FAME productivity, supporting surface-engineered SDRs as an alternative intensified reactor for short-contact-time continuous biodiesel production. Nevertheless, industrial-scale application requires further evaluation of hydrodynamic similarity, film stability at higher throughput, total system energy demand, equipment complexity, and long-term durability of the textured disc surface.

CRedit authorship contribution statement

Jatupon Yenchan: Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Sompong Putivisutisak:** Writing – review & editing, Validation, Methodology, Formal analysis. **Panut Bumphenkiattikul:** Writing – review & editing, Validation, Software, Investigation. **Sasiradee Jantasee:** Writing – review & editing, Writing – original draft, Methodology. **Chaiyan Chaiya:** Writing – review & editing, Supervision. **Puttiporn Thiamsinsangwon:** Writing – review & editing, Investigation. **La-ongthip Yuwansri:** Writing – review & editing, Investigation. **Kanokwan Ngaosuwan:** Writing – review & editing, Validation. **Suttichai Assabumrungrat:** Writing – review & editing, Visualization, Funding acquisition, Conceptualization. **Weerinda Mens:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cep.2026.110957.

Data availability

Data will be made available on request.

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